

Urysohn's Lemma

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History

In 1925, Pavel Urysohn published a paper which proved Urysohn's Metrization Theorem which states that every second countable T_3 space is metrizable. **Urysohn's Lemma** was included in the paper as a lemma he used to prove Urysohn's Metrization Theorem.

But **Urysohn's Lemma** is a important theorem in its own right.
"1st deep theorem in elementary topology" according to Munkres.

Countability

Definition

A set X is **countable** if there exists a injection from $X \hookrightarrow \mathbb{N}$.

Theorem

$\mathbb{Q}$ is countable.

Completeness Axiom

Definition

Suppose X is a subset of $\mathbb{R}$. If $\exists m \in \mathbb{R} \ni m \leq x, \forall x \in X$. Then X has a greatest lower bound in $\mathbb{R}$. If $\exists n \in \mathbb{R} \ni n \geq x, \forall x \in X$. Then X has a least upper bound in $\mathbb{R}$.

Open sets

Theorem

U is a open set $\iff \forall p \in U$, there exists a open set V_p such that $p \in V_p \subseteq U$.

Basis and Subbasis

Definition

Let $\mathcal{B}$ be a collection of subsets of X such that $\mathcal{B} \subseteq \mathcal{T}$. $\mathcal{B}$ is a **basis** set of $x \iff \forall U \in \mathcal{T}$, U is a arbitrary union of elements of $\mathcal{B}$.

Definition

Let $\mathcal{S}$ is a collection of subsets of X . $\mathcal{S}$ is a subbasis $\iff$ the set of all finite intersections is a basis set of X .

Normal Spaces

Definition

A space $(X, \mathcal{T})$ is normal $\iff$ for all disjoint closed sets A and B ,
 $\exists U, V \in \mathcal{T} \ni A \in U, B \in V$ and $U \cap V = \emptyset$.

Normal Spaces

Theorem (Normal Containment Theorem)

A space $(X, \mathcal{T})$ is normal $\iff$ for all open sets U and for all closed sets A such that $A \subseteq U$, there exists a open set V such that $A \subseteq V \subseteq \overline{V} \subseteq U$.

Proof.

$\implies$: Suppose a space $(X, \mathcal{T})$ is normal. For any open set U and closed set contained in A , $B = U^c$ is closed set which is disjoint from A . By normality $\exists V, W$ open sets such that $A \subseteq V$, $B \subseteq W$, and $V \cap W = \emptyset$. $U^c = B \subseteq W \implies W^c \subseteq B^c = U$.

$V \cap W = \emptyset \implies V \subseteq W^c$. Let $W' = \text{Int}(W^c)$. By definition of interior and closure, $A \subseteq V \subseteq W' \subseteq \overline{W'} \subseteq W^c \subseteq U$ □

Normal Spaces

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Proof.

$\Leftarrow$: Suppose X is a topological space where for all open sets U and for all closed sets A such that $A \subseteq U$, there exists a open set V such that $A \subseteq V \subseteq \overline{V} \subseteq U$. For all disjoint closed sets A and B , $A \subseteq B^c$ and $B \subseteq A^c$. $\exists U$ is a open set such that $A \subseteq U \subseteq \overline{U} \subseteq B^c$. $X - \overline{U}$ is open and disjoint from U and $B \subseteq X - \overline{U}$. Thus $A \subseteq U$ and $B \subseteq X - \overline{U}$ and $U \cap (X - \overline{U}) = \emptyset$, thus X is normal. $\square$

Continuous Functions

Definition

Let X and Y be topological spaces. $f : X \rightarrow Y$ is continuous $\iff$ for a open set U in Y , $f^{-1}(U)$ is a open set in X

Theorem (Subbasis Criterion for Continuity)

Let X and Y be topological spaces. Let $\mathcal{S}$ be a subbasis for Y . $f : X \rightarrow Y$ is continuous $\iff \forall A \in \mathcal{S}$, $f^{-1}(A)$ is open in X .

Urysohn's Lemma Statement

Theorem (Urysohn's Lemma Statement)

Let X be a topological space. X is a normal space $\iff$ for all disjoint closed sets A and B , $\exists f : X \rightarrow [0, 1]$ a continuous function such that $f(A) = 0$ and $f(B) = 1$.

Proving the backward direction

Proof.

$\Leftarrow$: Suppose X is a topological space where for all disjoint closed sets A and B , $\exists f : X \rightarrow [0, 1]$ a continuous function such that $f(A) = 0$ and $f(B) = 1$. $[0, \frac{1}{2})$ and $(\frac{1}{2}, 1]$ are disjoint open sets in $[0, 1]$. By definition of continuity, $U = f^{-1}([0, \frac{1}{2}))$ and $V = f^{-1}((\frac{1}{2}, 1])$ are open. They are also disjoint, since $[0, \frac{1}{2})$ and $(\frac{1}{2}, 1]$ are disjoint. Since $f(A) = 0$ and $f(B) = 1$, $A \subseteq f^{-1}([0, \frac{1}{2})) = U$ and $B \subseteq f^{-1}((\frac{1}{2}, 1]) = V$. Thus there exists open sets U and V , such that $A \subseteq U$ and $B \subseteq V$ and $U \cap V = \emptyset$. Thus X is normal. $\square$

The Idea Behind The proof

The forward direction is the more interesting (and more complicated) one. For this direction we have to *construct* a continuous function from scratch. To do this we want to create a contour of our space X with open sets and then make the contour finer and finer. Then we will use this contour to construct our function f .

The Lemma to Urysohn's Lemma

Theorem (Contour Creation Lemma)

Let X be a normal space. For any disjoint closed sets A and B . Then for each rational $r \in [0, 1]$ there exists a U_r such that $A \subseteq U_r$ and $B \subseteq (X - U_r)$ and for $r < s$, $\overline{U_r} \subseteq U_s$

The Lemma to Urysohn's Lemma

Proof.

Fix any pair of disjoint closed subsets of X , A and B . Let $P = \mathbb{Q} \cap [0, 1]$. Since P is countable and infinite, we will enumerate it as $P = \{p_n \mid n \in \mathbb{N}\}$ and for convenience $p_0 = 1$ and $p_1 = 0$. $\forall n \in \mathbb{N}$, $P_n = \{p_0, \dots, p_n\}$. We will use P to inductively construct a collection of open sets $\{U_p \mid p \in P\}$ which satisfies the theorem, or in other words a collection which satisfies the following property:

$$p < q \implies \overline{U_p} \subseteq U_q \text{ and } A \subseteq U_p, B \subseteq X - U_p (*)$$

The Lemma to Urysohn's Lemma

Proof.

Base Case: By definition of normal spaces, $\exists U_1, V$ open sets such that $A \subseteq U_1$ and $B \subseteq V$ and $U_1 \cap V = \emptyset$. Thus $B \subseteq X - U_1$. By the **Normal Containment Theorem**, $\exists U_0$ a open set such that $A \subseteq U_0 \subseteq \overline{U_0} \subseteq U_1$. $B \subseteq X - U_1 \subseteq X - U_0$. Thus $\{U_0, U_1\}$ satisfies property (*). □

The Lemma to Urysohn's Lemma

Proof.

Inductive Hypothesis: Assume that $\{U_p \mid p \in P_n\}$ satisfies property (*).

Inductive Step: $P_{n+1} = P_n \cup \{p_{n+1}\}$ thus $\{U_p \mid p \in P_{n+1}\} = \{U_p \mid p \in P_n\} \cup \{U_{p_{n+1}}\}$. Let $r = p_{n+1}$. Thus it suffices to prove that $\exists U_r, \forall s \in P_n \ni s < r, \overline{U_s} \subseteq U_r$ and $\forall t \in P_n \ni t > r, \overline{U_r} \subseteq U_t$. Note since $p_0 = 1$ and $p_1 = 0$, $0 < r < 1$. Thus $S = \{s \in P_n \mid s < r\}$ is not empty and has a max since P_n is finite. Similarly $T = \{t \in P_n \mid t > r\}$ is not empty and has a min since P_n is finite. Let $m = \max \{s \in P_n \mid s < r\}$ and $n = \min \{t \in P_n \mid t > r\}$.

The Lemma to Urysohn's Lemma

Proof.

Since $m < r$ and $r < n$, $m < n$. By the inductive hypothesis, $\overline{U_m} \subseteq U_n$. By the **Normal Containment Theorem**, $\exists U_r$ a open set such that $\overline{U_m} \subseteq U_r \subseteq \overline{U_r} \subseteq U_n$. $\forall s \in S$, $s = m$ or $s \neq m$. If $s = m$, $\overline{U_s} = \overline{U_m} \subseteq U_r$. If $s \neq m$, $s < m \implies \overline{U_s} \subseteq U_m \subseteq U_r$, by the inductive hypothesis. $\forall t \in T$, $t = n$ or $t \neq n$. If $t = n$, $\overline{U_r} \subseteq U_t = U_n$. If $t \neq n \implies t > n$, $\overline{U_r} \subseteq U_n \subseteq \overline{U_n} \subseteq U_t$, by the inductive hypothesis. Thus $A \subseteq U_0 \subseteq U_r$ and $U_r \subseteq U_1$, thus $B \subseteq X - U_r$. Thus condition (*) is satisfied for $\{U_p \mid p \in P_{n+1}\}$

Thus by induction $\{U_p \mid p \in P\}$ satisfies the condition (*) and the theorem is true. □

Exploiting the Contour

Let's now use the Contour Creation Lemma to construct a continuous function and finish proving Urysohn's Lemma.

Proof.

$\implies$: Suppose X is a normal space. Fix any two disjoint closed sets A and B . By the **Contour Creation Lemma**, $\exists \mathcal{U} = \{U_r\}_{r \in P}$ a collection of open sets which satisfy property $(*)$. For all $x \in X$, let $Q(x) = \{r \in P \mid x \in U_r\}$. Note if $x \notin U_1$, then $Q(x) = \emptyset$. Note if $x \in U_1$, 0 is a lower bound for $Q(x)$ since $Q(x) \subseteq [0, 1] \cap \mathbb{Q}$. Thus by Completeness of $\mathbb{R}$, $\inf Q(x)$ exists. Thus construct a function $f : X \rightarrow [0, 1]$, such that $\forall x \in X$:

$$f(x) := \begin{cases} \inf Q(x) & \text{if } x \in U_1 \\ 1 & \text{if } x \notin U_1 \end{cases}$$

Exploiting the Contour

Lemma (Lemma A)

If $x \in \overline{U}_r$, then $f(x) \leq r$.

Proof.

Suppose $x \in \overline{U}_r$, then $\forall s > r$, $x \in \overline{U}_r \subseteq U_s$, thus $s \in Q(x)$. Thus $(r, 1] \subseteq Q(x)$. Then by definition of infimum, $f(x) = \inf Q(x) \leq r$. □

Exploiting the Contour

Lemma (Lemma B)

If $x \notin U_r$, then $f(x) \geq r$.

Proof.

If $x \notin U_r$, then $x \notin U_s$ for all $s < r$ since $\overline{U_s} \subseteq U_r$. Thus r is a lower bound for $Q(x)$. Thus $r \leq \inf(Q) = f(x)$ by definition of infimum. □

Getting Continuity

Proof.

To prove continuity of f , we will use a suitable subbasis for $[0, 1]$ under the subspace topology of the normal topology on $\mathbb{R}$ and show that the preimage of subbasis elements are open.

$$\{[0, r) \mid r \in (0, 1]\} \cup \{(r, 1] \mid r \in [0, r)\}$$

is a subbasis of $[0, 1]$

Proving Continuity

Proof.

$\forall r \in (0, 1]$, to show $f^{-1}([0, r))$ is open we want to show that $\exists U_x$ a open set for all $x \in f^{-1}([0, r))$ such that $x \in U_x \subseteq f^{-1}([0, r))$. Thus $\forall x \in f^{-1}([0, r)) \implies f(x) \in [0, r)$. By density of rational in reals, $\exists a \in \mathbb{Q} \ni 0 \leq f(x) < a < r$. By the contrapositive of Lemma B, since $f(x) < a \implies x \in U_a$. Furthermore, by Lemma A $f(U_a) \subseteq f(\overline{U_a}) \subseteq [0, a] \subseteq [0, r) \implies U_a \subseteq f^{-1}([0, r))$. Thus $f^{-1}([0, r))$ is open.

Proving Continuity

Proof.

$\forall r \in [r, 1)$, this is symmetric to the argument above.

$\forall x \in f^{-1}((r, 1]) \implies f(x) \in [r, 1)$. By density of rational in reals,

$\exists a \in \mathbb{Q} \ni r < a < f(x) \leq 1$. By contrapositive of Lemma A,

$f(x) > a \implies x \notin \overline{U_a} \implies x \in X - \overline{U_a}$. Furthermore,

$f(X - \overline{U_a}) \subseteq f(X - U_a) \subseteq [a, 1] \subseteq (r, 1] \implies x \in X - \overline{U_a} \subseteq f^{-1}((r, 1])$. Thus $f^{-1}((r, 1])$ is open.

Thus f is continuous. Since $A \in U_0$, $\forall a \in A$, $f(a) \leq 0$ but since 0 is a lower bound for $Q(a)$, $0 \leq f(a) \leq 0 \implies f(a) = 0$. $\forall b \in B$, $b \notin U_1 \implies f(b) = 1$. Thus $f(A) = 0$ and $f(B) = 1$. □